ANSWER KEY

25900462 **Reg. No :……**
Name :…….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCIAM200 - EMBEDDED SYSTEM PROGRAMMING TOOLS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt a t least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each Question Carries 1 Marks

1. b) 8 [K]/[CO1]
2. b) PIL / Pillow [K]/[CO2]
3. b) grid [U]/[CO3]
4. b) x is a string and y is a floating-point number [U]/[CO1]
5. a) To perform arithmetic operations [U]/[CO2]
6. a) C [K]/[CO3]
7. b) To easily represent numbers in different bases like binary, octal, or hex [U]/[CO2]
8. b) 3 [K]/[CO3]
9. b) To enable remote access and internet connectivity [U]/[CO1]
10. c) place [U]/[CO3]

Part B
Short Answer Questions
Answer any questions
Each Questions Carries 5 Marks

11. Main Numerical Data Types and Sequence Types in Python [K]/[CO1]

Type	Category	Example & Description
int	Numerical	x = 10 → Integer – whole number
float	Numerical	y = 3.14 → Floating-point (decimal) number
complex	Numerical	z = 2 + 3j → Complex number with real and imaginary parts

str	Sequence	name = 'Python' → String – sequence of characters
list	Sequence	nums = [1, 2, 3] → Ordered, mutable collection
tuple	Sequence	t = (4, 5, 6) → Ordered, immutable collection
range	Sequence	r = range(5) → Sequence of numbers used in loops

12. Program to Perform Digital Read from a GPIO Pin [U]/[CO2]

```
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setup(17, GPIO.IN)     # GPIO 17 as input (push button)

button_state = GPIO.input(17)
print("Button State:", button_state)
GPIO.cleanup()
```

Explanation: GPIO.input(pin) returns 1 (HIGH) when the button is pressed and 0 (LOW) when released.

13. Which Method is Used to Start the Tkinter Event Loop? [K]/[CO3]

Answer: mainloop()

```
from tkinter import *
root = Tk()
root.mainloop()
```

Purpose: Keeps the window open and waits for user interaction.

14. Purpose of GPIO.setup() and GPIO.cleanup() [U]/[CO2]

GPIO.setup(pin, mode): Defines the pin mode as input or output.
GPIO.cleanup(): Frees GPIO resources and resets pins.

```
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setup(18, GPIO.OUT)
GPIO.output(18, True)
GPIO.cleanup()
```

Explanation: Prevents warnings or errors in the next program run.

15. What Does GUI Stand For? [K]/[CO3]

Answer: Graphical User Interface
A GUI allows users to interact with software using windows, buttons, and icons instead of command-line text.

16. Modes of GPIO Operation and Their Difference [k]/[CO1]

Mode	Purpose	Example
Input	Reads data/signals from sensors or buttons.	GPIO.setup(pin, GPIO.IN)
Output	Sends signals to devices like LEDs or motors.	GPIO.setup(pin, GPIO.OUT)

Difference: Input mode receives signals, while Output mode sends signals.

Part C

Essay Questions

Answer any 2 Questions

Each Questions Carries 10 Marks

17. Port Layout of Raspberry Pi 4 [U / CO1]

Main Points:

- The Raspberry Pi 4 has multiple ports and connectors for interfacing, display, and power.
- Key Components and their Roles:
  • USB Ports – 2 × USB 3.0 (high-speed), 2 × USB 2.0 (keyboard, mouse, pen drive)
  • HDMI Ports – Dual micro-HDMI outputs, supports dual monitors up to 4K
  • Ethernet Port – Provides wired network connectivity (Gigabit Ethernet)
  • GPIO Pins – 40-pin header for sensors, LEDs, actuators (Input/Output modes)
  • Power Port – USB-C connector (5V/3A supply)
  • Audio/Video Jack – For sound and analog video output

• Camera (CSI) and Display (DSI) Ports – For camera and LCD connections
- Purpose: Enables hardware interfacing, multimedia, and internet connectivity.

18. Working of Label, Button, Entry, and Checkbutton Widgets [K / CO3]

Widget	Purpose	Example Code	Description
Label	Display text or images	Label(root, text="Hello").pack()	Non-interactive text display
Button	Perform an action when clicked	Button(root, text="Click", command=func).pack()	Executes a function on click
Entry	Single-line text input	Entry(root).pack()	Used for user data input
Checkbutton	Toggle between two states (On/Off)	Checkbutton(root, text="Accept").pack()	Stores True/False value

Each widget works under the Tkinter GUI system and responds to user actions when the event loop (mainloop()) is active.

19. Basic Flow of a Python Program (Input → Processing → Output) [U / CO1]

Main Points:
1. Input – Collect data from the user or an external source.
   Example: num = int(input("Enter a number: "))
2. Processing – Perform computation or logic.
   Example: square = num ** 2
3. Output – Display or store the result.
   Example: print("Square is:", square)

Complete Example:

```
num = int(input("Enter a number: "))
square = num ** 2
print("Square is:", square)
```

Flow: User enters → program processes → displays result. Demonstrates sequential execution in Python.

20. Control Statements in Python (Decision-making & Iteration) [U]/[CO2]

Main Points:
- Control statements guide program execution flow.
- Types:
  • Decision-making: if, elif, else
  • Iteration: for, while loops

Example Program:

```
for i in range(1, 6):
    if i % 2 == 0:
        print(i, "is even")
    else:
        print(i, "is odd")
```

Explanation:
- The for loop repeats for values from 1 to 5.
- The if statement checks even/odd condition.

Output:
1 is odd
2 is even
3 is odd
4 is even
5 is odd